

PROGRAMMING IN C (HDOC DAY-4)

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Problem-

to find the product of two integers without using the multiplication operator *.

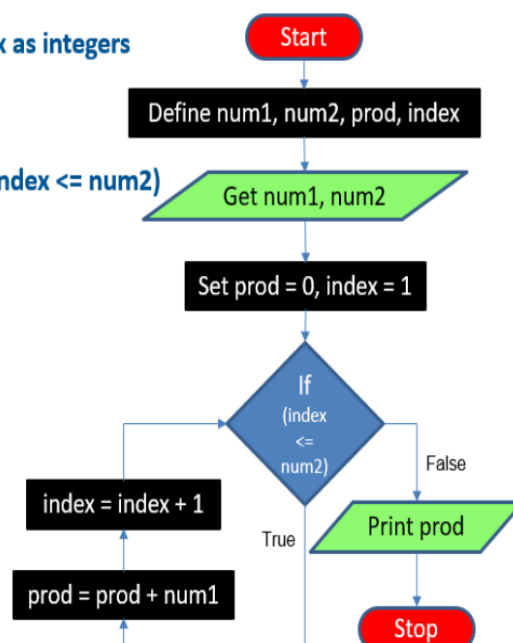
Tasks-

- Identify the mistakes in the given algorithm.
- Rewrite the corrected algorithm.
- Identify the mistakes in the given flowchart.
- Redraw the corrected flowchart.
- Verify your corrected algorithm and flow chart using the given test cases.

Given Algorithm and Flow Chart

- **Algorithm & Flowchart**

1. Define num1, num2, prod, index as integers
2. Get num1, num2
3. Set prod = 0, index = 1
4. Repeat steps 4.1 and 4.2 until (index <= num2)
 - 4.1. prod = prod + num1
 - 4.2. index = index + 1
5. Print prod
6. Stop



Mistake Identification in Algorithm-

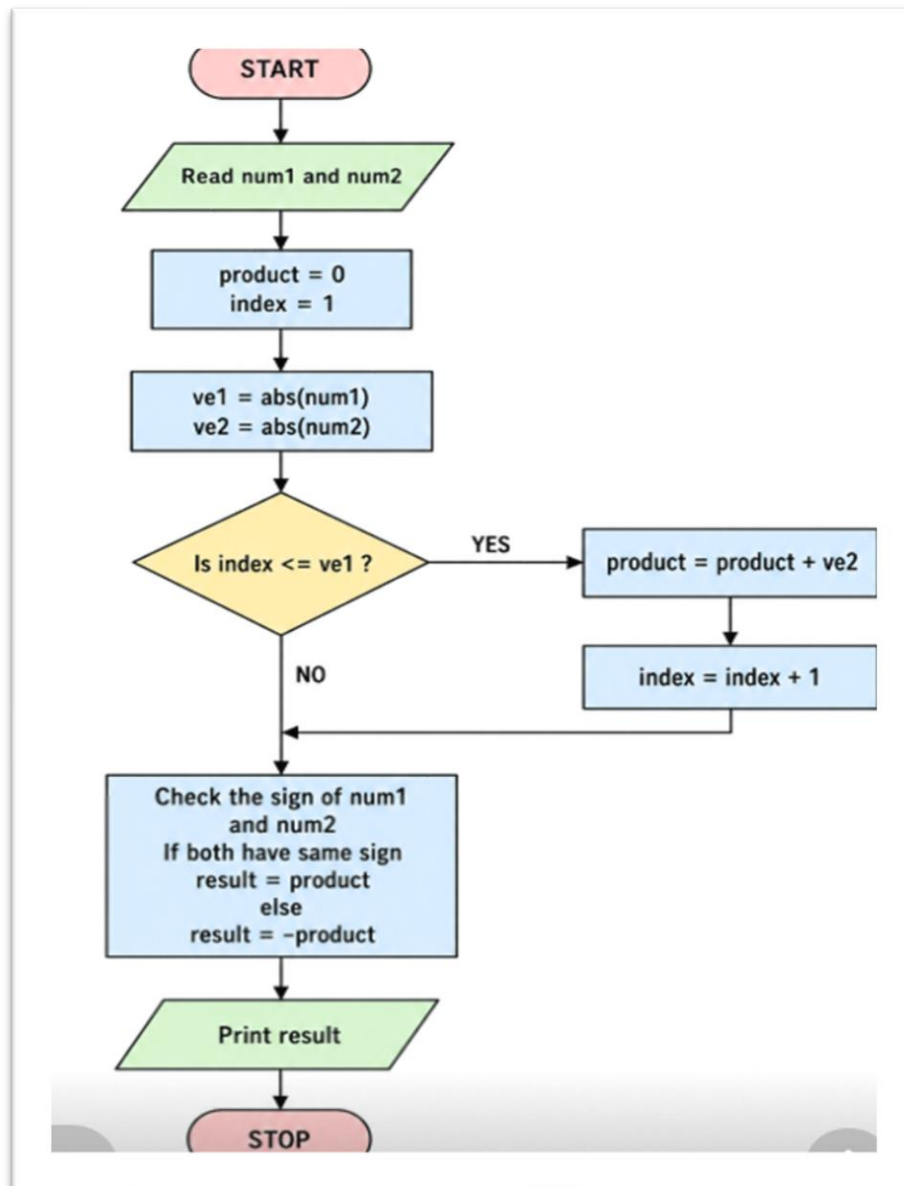
- This algorithm will definitely work for positive numbers but in case of negative numbers there will be a logical error.
- If one of the number to be multiplied is zero then it will not handle multiplication correctly.

Correct Algorithm

1. Start.
2. Get num1 and num2.
3. Initialise product=0 and index=1.
4. Convert both numbers to their absolute values.
5. Repeat the step 5.1 and 5.2 till(index<=ve1)
 - 5.1. product=product+ve2;
 - 5.2. index=index+1;
6. Print the product, check the sign and print product with desired sign.
7. Stop.

PROGRAM(DAY 4 PROBLEM)

Correct Flowchart:



PROGRAM(DAY 4 PROBLEM)

```
1  #include <stdio.h>
2  #include <stdlib.h>
3
4  int main()
5  {
6  int num1,num2;
7  int product=0;
8  scanf("%d%d",&num1,&num2);
9  int v1=(int)abs(num1);
10 int v2=(int)abs(num2);
11 for(int i=1;i<=v2;i++)
12 {
13     product+=v1;
14 }
15 if((num1<0 && num2>0) || (num1>0 && num2<0))
16 {
17     product=-product;
18 }
19 printf("%d",product);
20 return 0;
21 }
```

Terminal Window(Screenshots):

```
-6
5
-30
PS D:\git hub\HDOC\hdocday4>
```

```
12
3
36
```

```
-1
-9
9
PS D:\git hub\HDOC\hdocday4>
```